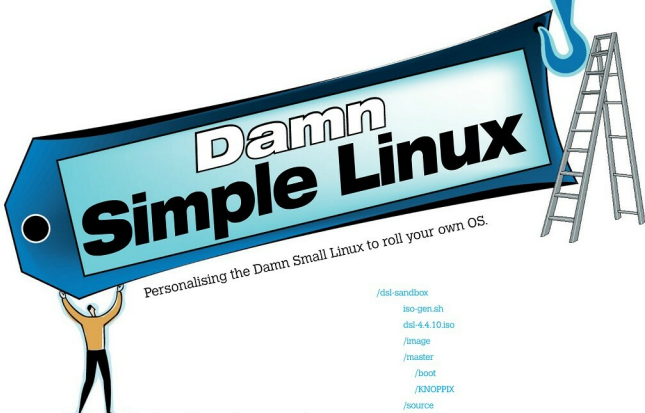




There's an old unused computer at home that doesn't have a great hardware profile—a Pentium II processor with 64 MB of RAM. I wanted to use this computer meaningfully, and hence came up with this idea of customising Damn Small Linux (DSL), a small GNU/Linux distribution that can work with minimal hardware.

The users of this system were my parents, who know very little about computers and merely wanted a simple interface to browse the Web and check e-mails. I decided to make the system as simple as possible so that anyone could switch it on and start using it.

Here are the requirements specifications:

- 1 Use low-profile hardware
- 2 Quick booting
- 3 Show my name during start up :-)
- 4 Should not require entering a user name and password to log in to the system
- 5 Show the Web browser with some URL, as default

Setting up the environment

Let's get started with the actual work. I am using Debian Lenny (5.0.3) as my base system to carry out these customisations (you may require root permissions in order to carry out most of the tasks explained; I use *sudo* to achieve this). Download the DSL 4.4.10 ISO image from www.damnsmalllinux.org/download.html. This is the ISO image you are about to customise for your needs. Create a new folder in your system with the following layout:

```
/dsl-sandbox
  iso-gen.sh
  dsl-4.4.10.iso
  /image
  /master
  /boot
  /KNOPPIX
  /source
  /KNOPPIX
```

You have an *iso-gen.sh* script—which you can use to generate your final ISO image—in the above folder layout. Its contents are as follows:

```
#!/bin/sh
# iso-gen.sh: Generate the final iso.

cd master
rm -f KNOPPIX/md5sums; find -type f -exec md5sum {} \; >> KNOPPIX/
md5sums
mkisofs -pad -l -r -J -V "KNOPPIX" -no-emul-boot -boot-load-size 4 \
-boot-info-table -b boot/isolinux/isolinux.bin -c boot/isolinux/boot.cat \
-o ./styleisen.iso.
```

You need to mount the DSL ISO image that you have downloaded in order to have access to all the files that it contains. The steps are illustrated below. Note that you'll require the *cloop-utils* package installed in your Debian system—it offers the following *create_compressed_fs* and *extract_compressed_fs* tools.

```
$ cd dsl-sandbox
$ sudo losetup /dev/loop0 dsl-4.4.10.iso
$ sudo mount /dev/loop0 image
$ cd image; sudo find . -size -1000k -type f -exec cp -p --parents {} /master/ \;
$ cd ..
$ sudo cp -pR image/boot/* master/boot/
$ umount image
$ sudo losetup -d /dev/loop0
```